

Better Executives Work at Better Firms: Identifying Talent Sorting from Mobility Networks

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The economic question

Does executive assignment matter for aggregate output?

Misallocation lowers aggregate productivity (Hsieh and Klenow 2009)

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Executives are few, but their decisions scale with the firm.

Who runs which firm is an aggregate productivity question.

A simple assignment model

Firm i combines labor L with two fixed factors: productivity A_i and executive talent Z_m (Lucas 1978). At wage w , it chooses labor to maximize profit:

$$\pi_{im} = \max_L \left\{ A_i Z_m^\nu L^{1-\nu} - wL \right\}.$$

At optimal labor, revenue is Cobb–Douglas in the *two fixed factors*:

$$Y_{im} = c(w) A_i^{1/\nu} Z_m.$$

Complementarity means assignment matters

$$\frac{\partial^2 Y}{\partial A \partial Z} > 0$$

The efficient allocation matches better executives to better firms (Gabaix and Landier 2008; Terviö 2008).

Sorting raises expected output

With $a = \ln A/\nu + \ln c(w)$ and $z = \ln Z$ jointly normal, expected revenue per match is

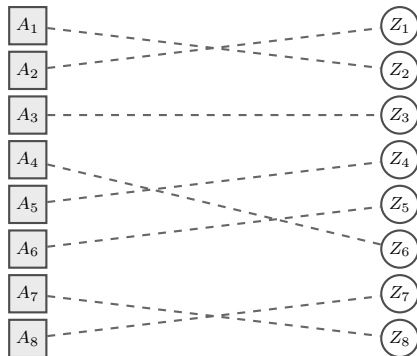
$$\mathbb{E}[Y_{im}] = \exp \left\{ \mu_a + \mu_z + \frac{1}{2}(\sigma_a^2 + \sigma_z^2) + \text{Cov}(a, z) \right\}.$$

Holding marginal distributions fixed, expected aggregate revenue increases in $\text{Cov}(a, z)$.

Talent allocation in one picture

firms, ranked

executives, ranked



How far is the economy from Positive Assortative Matching?

Measuring sorting from mobility

How can we measure executive talent?

- 1 Surveys and diaries record what executives do (Bandiera et al 2020)
- 2 Education, experience, and management style provide observable proxies
- 3 Outcomes reveal persistent contributions when executives move (Bertrand and Schoar 2003, Abowd, Kramarz and Margolis 1999)

The Hungarian executive mobility network, 1990-2022

The graph links firms to executives who served them. Outcomes record firm performance during executive spells.

	Value
Executives	1,304,580
Firms	1,036,231
Unique firm–executive links	1,937,552
Links in giant component	823,158 (42.5%)
Connected components	514,085
Bridge share of edges	81.8%
Mean shortest path between executives (giant)	19.5

What is new

- 1 Follow chains in the **mobility network** connecting firms and executives that never meet
- 2 Model sorting with four parameters instead of estimating millions of fixed effects
- 3 Use every retained component, including the tree-like parts discarded by leave-out methods

What we find

Sorting is high:

$$\rho = 0.52\text{--}0.55.$$

Random assignment lowers expected revenue by **12.9%**.

Why fixed effects lose most of the economy

A minimal model of outcomes

Firm performance under executive m is

$$y_{im} = a_i + z_m + \varepsilon_{im}.$$

Firm fundamentals a_i and executive talent z_m are latent, observed only through outcomes.

Mobility reveals talent

Measure executive talent z_m from switches within the firm: $\Delta \hat{z}_m \approx \Delta y_{im}$.

Treat a_i , z_m as *parameters* and run OLS with two sets of dummies:

Labor: (Abowd, Kramarz and Margolis 1999; Andrews et al 2008; Kline, Saggio and Sølvesten 2020).

Executives: (Bertrand and Schoar 2003; Fenizia 2022; Metcalfe, Sollaci and Syverson 2023)

Problem 1: fixed effects need a connected component

Firm and executive effects are identified only up to a component-specific normalization.

Cannot compare their levels across disconnected components.

Standard practice: keep the largest component, drop the rest.

Problem 2: noise biases second moments

Because of ε — not lack of mobility — estimated effects are noisy.

Second moments are biased: $\widehat{\text{Var}}$ too big, $\widehat{\text{Cov}}$ too small (Andrews et al 2008; Kline, Saggio and Sølvesten 2020; Koren, Orbán and Telegdy 2025)

Limited mobility makes both problems worse.

Both problems are worse for executives

Executive mobility is rare: the graph is very sparse.

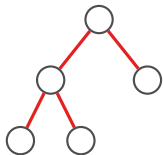
Hundreds of thousands of disconnected components.

Firm outcomes are noisy.

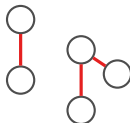
Estimates are especially noisy — *and* the usual corrections do not apply.

Trees, forests, cycles, and bridges

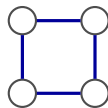
a tree:
no cycles



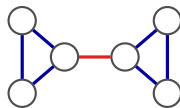
a forest:
disjoint trees



a cycle:
multiple paths between nodes



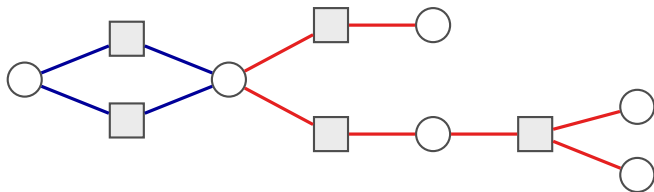
a bridge:
removing it disconnects



A **bridge** is an edge whose removal disconnects the graph.

On a tree, every edge is a bridge.

Leave-out corrections need cycles



blue: on a cycle, leverage < 1 — usable

red: bridges, leverage $= 1$ — dropped

Squares are firms, circles are executives, edges are jobs.

Leave-out estimation keeps one in ten observations

Stage	Matched pairs	%
All unique firm–executive pairs	1,732,154	100.0%
Keep giant component	823,135	47.5%
Drop bridges (leverage = 1)	265,995	15.4%
Keep largest leave-out-connected block	175,661	10.1%

Giant component is not representative

	Giant	Other
Employment, mean	9.1	3.5
Log revenue, mean	9.37	8.65
Manufacturing	8.5%	9.5%
Founder managed	46.8%	72.1%

Identification from mobility chains

Sorting on an edge

For an isolated edge — the building block of a tree —

$$\begin{pmatrix} a_i \\ z_m \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \sigma_a^2 & \rho\sigma_a\sigma_z \\ \rho\sigma_a\sigma_z & \sigma_z^2 \end{bmatrix}\right).$$

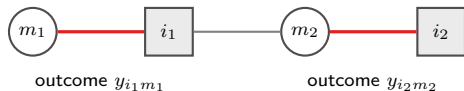
Here ρ is the local edge correlation.

Four parameters describe sorting

$$\boldsymbol{\theta} = (\sigma_a, \sigma_z, \rho, \sigma_\varepsilon), \quad \varepsilon_{im} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

ρ measures sorting. The three scale parameters capture firm, executive, and match heterogeneity.

Covariance without a common match shock



The endpoint nodes m_1 and i_2 are $h = 3$ edges apart; the outcomes share no match shock.

Covariance from sorting:

$$\rho(\sigma_a + \rho\sigma_z)(\sigma_z + \rho\sigma_a).$$

Covariance decays along the mobility tree

For endpoint nodes h edges apart:

$$\text{Cov}(P - F - \dots - F - P) = \rho^{h-2} (\sigma_a + \rho\sigma_z)^2$$

$$\text{Cov}(F - P - \dots - P - F) = \rho^{h-2} (\rho\sigma_a + \sigma_z)^2$$

$$\text{Cov}(P - F - \dots - P - F) = \rho^{h-2} (\sigma_a + \rho\sigma_z)(\rho\sigma_a + \sigma_z)$$

The **level** mixes σ_a , σ_z , ρ .

Two additional hops multiply covariance by ρ^2 .

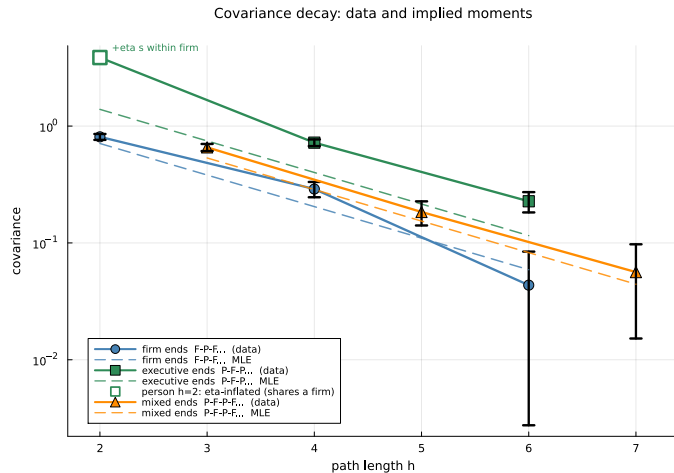
Correlation decay along the family tree

Clark writes for most relatives (Fisher 1918; Gimelfarb 1981; Clark 2023)

$$\rho_n = h^2 b^n, \quad b = \frac{1 + m}{2}.$$

Here ρ_n is status correlation at genealogical distance n , h^2 is heritability, and m is spouse correlation in genetic values.

Covariance decay in the Hungarian data



What about within-firm shocks over time?

$$\text{Cov}(P - F - P) = (\sigma_a + \rho\sigma_z)^2$$

only if ε_{im} is i.i.d. over time. For autocorrelated shocks, biased upward.

But common shocks are **localized**. Sorting is identified by the **speed of decay over long chains**.

Three ways to separate sorting from firm shocks

- 1 **Minimum distance:** drop the contaminated same-firm moment
- 2 **Firm average:** allow arbitrary dependence within a firm
- 3 **AR(1) match shocks:** estimate persistence within the firm

A model for sparse mobility networks

The full network model

Collect every latent firm and executive type in $\mathbf{x}$:

$$\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^{-1}).$$

$\mathbf{Q}$ is sparse: directly linked firms and executives are conditionally related; absent links are zeros.

This is a **Gaussian Markov random field** (Rue and Held 2005).

The familiar case: an AR(1)

$x_t = \rho x_{t-1} + u_t$ on a line graph:

$$\mathbf{Q} \propto \begin{bmatrix} 1 & -\rho & & \\ -\rho & 1 + \rho^2 & -\rho & \\ & -\rho & 1 + \rho^2 & -\rho \\ & & -\rho & 1 \end{bmatrix}$$

Sparse precision, dense covariance

The MA representation of an AR(1) is dense:

$$\Sigma = \mathbf{Q}^{-1} \propto \begin{bmatrix} 1 & \rho & \rho^2 & \rho^3 \\ \rho & 1 & \rho & \rho^2 \\ \rho^2 & \rho & 1 & \rho \\ \rho^3 & \rho^2 & \rho & 1 \end{bmatrix}$$

Partial correlation: only with neighbors. Unconditional autocorrelation at lag d : ρ^d .

On a tree, dependence follows the unique path

$$\text{Cov}(x_k, x_\ell) = \sigma_k \sigma_\ell \rho^{d(k, \ell)}$$

Real networks have cycles

Multiple paths push covariances above ρ^d .

The GMRF handles cycles exactly, through $\mathbf{Q}^{-1}$.

Why observed links reveal sorting

Before matching, types are independent

Before links form,

$$a_i \sim \mathcal{N}(0, \omega_a^2), \quad z_m \sim \mathcal{N}(0, \omega_z^2).$$

Any correlation among linked firms and executives must come from selection into links.

Match or not: two payoffs

$$\text{match: } \kappa + \frac{a_i z_m}{\tau},$$

$$\text{no match: } \frac{1}{\tau} \left(\alpha_1 a_i + \zeta_1 z_m + \frac{\alpha_2}{2} a_i^2 + \frac{\zeta_2}{2} z_m^2 \right).$$

$\tau > 0$ scales match frictions. The match term $a_i z_m$ makes firm and executive types complements; the linear and quadratic terms make outside options type-dependent.

Their difference is the logit index

For short, define

$$\eta(a_i, z_m) := \kappa + \frac{1}{\tau} \left[a_i z_m - \alpha_1 a_i - \zeta_1 z_m - \frac{\alpha_2}{2} a_i^2 - \frac{\zeta_2}{2} z_m^2 \right].$$

Then the two states are

$$\Pr(L_{im} = 1 \mid a_i, z_m) = \frac{e^\eta}{1 + e^\eta}, \quad \Pr(L_{im} = 0 \mid a_i, z_m) = \frac{1}{1 + e^\eta}.$$

The likelihood of the observed graph

Joint density of types and links: prior $\times$ each edge that did or did not happen,

$$p(\mathbf{x}, E) \propto p_0(\mathbf{x}) \prod_{(i,m) \in E} \Pr(L_{im}=1 \mid \mathbf{x}) \prod_{(i,m) \notin E} \Pr(L_{im}=0 \mid \mathbf{x})$$

Links are rare events. When e^η is small,

$$\Pr(L_{im} = 1 \mid a_i, z_m) \simeq e^{\eta(a_i, z_m)} \quad \text{and absent pairs drop out.}$$

The **sparse-link approximation**: keep only the positive links.

The tilted distribution is exactly our GMRF

Each observed link adds a quadratic term to the exponent:

$$p_E^+(\mathbf{x}) \propto p_0(\mathbf{x}) \prod_{(i,m) \in E} e^{\eta(a_i, z_m)} \propto \exp\left(-\frac{1}{2}\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{b}^\top \mathbf{x}\right)$$

$$\mathbf{Q} = \underbrace{\mathbf{Q}_0}_{\text{indep. prior}} + \underbrace{\frac{1}{\tau} \mathbf{C} \mathbf{D}}_{\text{repeated selection}} - \underbrace{\frac{1}{\tau} \mathbf{A}}_{\text{complementarity}} = \frac{1}{1 - \rho^2} \mathbf{S} \left[(1 - \rho^2) \mathbf{I} + \rho^2 \mathbf{D} - \rho \mathbf{A} \right] \mathbf{S}$$

An equality, not an analogy: with $\tau = \sigma_a \sigma_z (1 - \rho^2) / \rho$, link formation *generates* the variance-stable precision.

When is the approximation valid?

The sparse-link approximation drops the shared logit denominators.

Valid when the graph is sparse and forest-like; dense cyclical regions invalidate it.

So we **prune** the few cycle-closing edges in the dense core — keeping every node and every component — and estimate on the nearly-forest graph.

Opposite of leave-out: KSS lives on the cycles, we lean on the trees. But we keep the data.

The likelihood function

$$\ell(\boldsymbol{\theta}) = -\frac{K}{2} \ln(2\pi) - \frac{1}{2} \ln \det \boldsymbol{\Omega} - \frac{1}{2} \mathbf{y}^\top \boldsymbol{\Omega}^{-1} \mathbf{y}$$

$$\boldsymbol{\Omega} = \mathbf{V} \mathbf{Q}^{-1} \mathbf{V}^\top + \sigma_\varepsilon^2 \mathbf{R} \quad (\mathbf{V} \text{ maps edges to types})$$

$$\mathbf{Q} = \frac{1}{1 - \rho^2} \mathbf{S} \left[(1 - \rho^2) \mathbf{I} + \rho^2 \mathbf{D} - \rho \mathbf{A} \right] \mathbf{S}, \quad \mathbf{S} = \text{diag}(\sigma_a^{-1} \mathbf{I}_{N_f}, \sigma_z^{-1} \mathbf{I}_{N_m})$$

$\mathbf{A}$ = adjacency, $\mathbf{D}$ = degrees, ρ on the edges, scales on the diagonal.

Which estimator, when?

Network	Estimator
Dense (much mobility)	leave-out (KSS) — little data lost
Sparse, forest-like	GMRF
In between	run both, compare

Sparsity is where the GMRF is useful — and where leave-out corrections discard most observations.

What the data say

Sorting is high: $\rho \approx 0.52\text{--}0.55$

Same graph, same outcomes (log real revenue, demeaned by year and industry):

Parameter	Naive MLE	Min. distance	Firm-average	AR(1) MLE
ρ	0.379	0.555	0.537	0.518
Implied corr. (a, z)	0.381	0.555	0.544	0.525

What does $\rho \approx 0.52$ mean?

On the pruned tree-like graph, standardized latent types have correlation

$$\rho^d$$

at network distance d .

Direct neighbors: 0.52. Two steps apart: 0.27. Four steps apart: 0.07.

Sorting is a large source of revenue dispersion

Component	Share
Firm effects	28.4%
Executive effects	1.4%
Sorting covariance	6.6%
Match noise (within-firm correlated)	63.5%

Could other similarities explain the pattern?

The threat: similarity, not propagation

Spells along a mobility path occur at different times.

Shared sector, region, or cohort could still produce similar outcomes: the homophily critique.

Conditioning on the path removes most correlation

We compare outcomes at the ends of 432,114 isolated shortest paths.

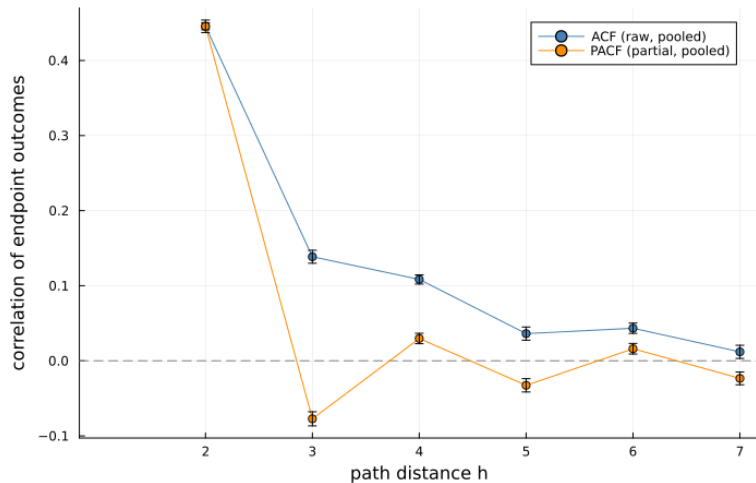
Raw endpoint correlations persist with distance. After controlling for every outcome between the endpoints, little remains:

$$|\text{PACF}| \leq 0.08 \quad \text{for } h \geq 3.$$

Markov pattern: conditional on the path, distant outcomes are nearly uncorrelated.

Preliminary pooled estimates; model-based inference comes next.

Raw correlation persists; partial correlation does not



At $h = 2$, there is nothing between the endpoints to condition on, so ACF and PACF coincide. For $h \geq 3$, PACF stays close to zero.

Disconnected components: small negative rejection

The GMRF predicts exactly zero covariance across disconnected components.

	Diagnostic	Estimate
Cross-component pairs		58,785
Covariance (SE)		-0.0629 (0.0208)
Correlation		≈ -0.013
Formal test	$\chi^2(10) = 21.38, p = 0.0186$	

Exact zero is rejected, but the estimate is tiny and negative—not the positive floor implied by a broad common factor.

Economic stakes

What is sorting worth?

Hold fixed:

- the marginal distributions of firm fundamentals and executive talent;
- match-noise dispersion;
- the number of matches.

Then change only how firms and executives are assigned.

This is a model-based, partial-equilibrium reassignment exercise.

Three counterfactuals

Scenario	Δ expected revenue
Random assignment ($\rho = 0$)	-12.9%
Compress talent dispersion, mean-preserving ($\sigma_z \rightarrow 0$)	-12.9%
Perfect sorting ($\rho = 1$)	$\approx +45\%$

Perfect sorting is an upper bound, not a policy.

Other sparse latent-type networks

Examples: buyer–supplier networks and international trade.

The graph need not be bipartite.

Conclusion

- 1 Sorting between executive talent and firm fundamentals is **high**: $\rho \approx 0.52\text{--}0.55$
- 2 Four parameters replace millions of fixed effects — using **every retained component and nearly all the data**, not 10%
- 3 Identification is transparent: covariance decay along mobility chains, and it is testable
- 4 Random assignment would cost **12.9%** of revenue

Appendix

The leave-one-out correction and leverage

Kline, Saggio and Sølvssten (2020) estimate each effect *without* observation i .

FE estimates are linear in outcomes; observation i enters its own fitted value with weight $h_i = \mathbf{leverage}$. Bias correction re-weights by $1/(1 - h_i)$.

The catch: when $h_i = 1$, the effect is estimable *only* from observation i . These observations are dropped.

$h_i = 1$ **exactly when edge i is a bridge.**

Preferred AR(1) parameter estimates

$$\hat{\sigma}_a = 1.08, \quad \hat{\sigma}_z = 0.24, \quad \hat{\sigma}_\varepsilon = 1.72, \quad \hat{\eta} = 0.51.$$

The correlated match shock absorbs persistent within-firm conditions.

The precision matrix, one edge at a time

Zoom in on a single firm–executive pair (degree 1 each):

$$\mathbf{Q}_{\{i,m\}} = \frac{1}{1-\rho^2} \begin{bmatrix} \sigma_a^{-2} & -\rho \sigma_a^{-1} \sigma_z^{-1} \\ -\rho \sigma_a^{-1} \sigma_z^{-1} & \sigma_z^{-2} \end{bmatrix} \iff \text{Var} \begin{pmatrix} a_i \\ z_m \end{pmatrix} = \begin{bmatrix} \sigma_a^2 & \rho \sigma_a \sigma_z \\ \rho \sigma_a \sigma_z & \sigma_z^2 \end{bmatrix}$$

Off-diagonal entries of $\mathbf{Q}$ are nonzero **only along observed edges**.

On a forest, every linked pair has correlation exactly ρ .

What is a Gaussian Markov random field?

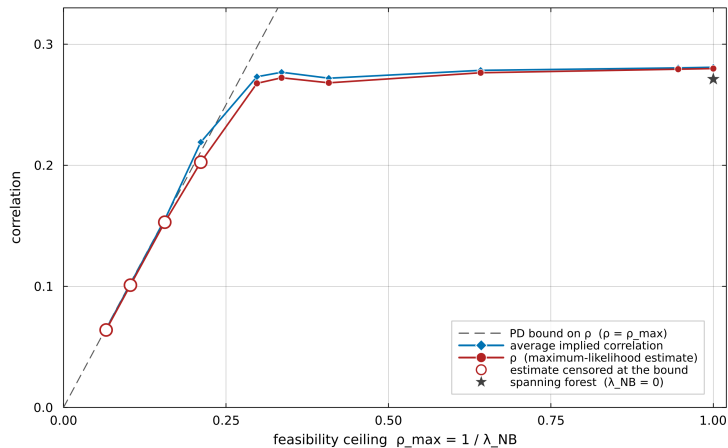
Jointly normal variables with a **sparse precision matrix**.

Zero in $\mathbf{Q} \Leftrightarrow$ conditional independence given everyone else.

Conditional dependence is local — only direct neighbors.

Unconditional dependence travels along paths.

Prune cycle-closing links, keep every component



We remove problematic cycle-closing links, while retaining every node and component.

Counterfactual formula

Expected revenue per match under the lognormal model:

$$\frac{Y}{n} = \exp \left\{ \mu_a + \mu_z + \mu_\varepsilon + \frac{1}{2} (V_a + V_z + V_\varepsilon + 2 \operatorname{Cov}(a, z)) \right\}$$

Hold the marginal distributions fixed, change the allocation, and read off the change in expected output.

Executive effects in the literature

Fenizia (2022): Italian public managers, AKM with the Andrews et al. (2008) correction. Only 24% of managers ever move, and the bias-corrected decomposition keeps 21% of observations (the largest connected set).

Metcalf, Sollaci and Syverson (2023): retail managers, *negative* estimated correlation.

Noise inflates the estimated variances and induces negative covariance between the two estimated effects, pushing their correlation below zero. (Mechanics in appendix.)

Our companion papers

Wohak and Koren (2025): leave-out on sparse graphs. KSS and the GMRF use *opposite* parts of the graph — cycles vs bridges. Even off the bridges, high leverage makes second-moment estimates noisy, and noise plus nonnegativity constraints means bias.

Koren, Orbán and Telegdy (2025): CEO value. A time-series estimator that removes spurious pre-trends from within-firm error correlation; after the correction, executive changes look unrelated to prior performance in the same Hungarian data.

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Why fixed-effects correlations come out negative

The FE estimates are linear combinations of outcomes, including the noise.

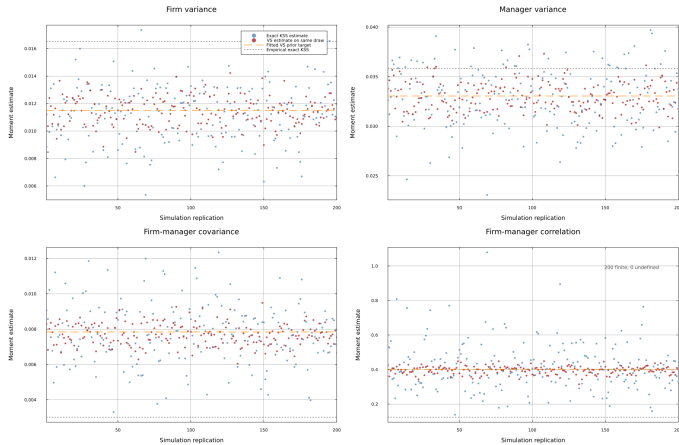
Whenever the firm effect loads *positively* on a noise term, the executive effect at the same match loads *negatively*: the two dummies share the same variation.

Inflated variances bias the correlation toward zero; the mechanical negative covariance pushes it further down.

Estimates in the literature are lower, sometimes negative.

Leave-out is noisy even where it exists

Finite-sample dispersion of KSS and VS estimates on the common graph



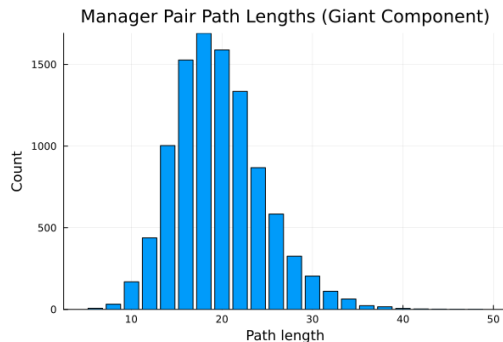
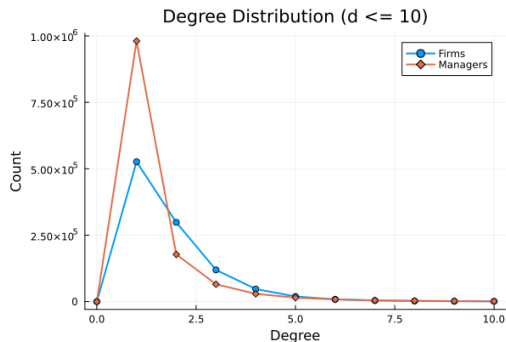
Simulations on the common graph: KSS second moments (blue) disperse far more than the GMRF (red).

Model fit: where log-revenue variance goes

Component	Share
Firm effects	28.4%
Executive effects	1.4%
Sorting covariance	6.6%
Match noise (within-firm correlated)	63.5%

The sorting covariance is several times the executive component.

The bipartite mobility graph



The giant component holds 42% of edges; the rest are mostly single firm–executive pairs.

Monte Carlo: correlated errors on the Hungarian graph

Hungarian common graph (K=91,085), 2000 reps/arm — truth $\rho = 0$

